Airline Flight operations Analytics Using Databricks | End-to-End Data Engineering Project



1. Introduction

Airline operations generate large amounts of data every day, including flight schedules, passenger details, delays, routes, and revenue.

In this project, I transformed raw airline flight data into meaningful business insights using Databricks, SQL, and Medallion Architecture.

2. Technology Stack

- Databricks
- SQL
- Medallion Architecture
- Databricks AI/BI Dashboards

3. Dataset Overview

The dataset contains 1,365 flight records covering January to December 2025.

It includes flight IDs, airline names, aircraft details, routes, passenger counts, ticket revenue, and flight delays.

4. Bronze Layer — Raw Data Ingestion

The Bronze layer stores raw source data with ingestion metadata.

SQL Query:

```
CREATE OR REFRESH STREAMING TABLE bronze_airline_flights
AS
SELECT
  *,
  _metadata.file_name AS source_file,
  current_timestamp() AS loaded_at
FROM STREAM read_files(
  '/Volumes/workspace/flights/raw_files/',
  format => 'csv',
  header => true
);
```

Verify Bronze Records:

```
SELECT COUNT(*) AS bronze_count
FROM workspace.flights.bronze_airline_flights;
```

Output:

MetricResultBronze records1,365

What this tells us: All 1,365 source records were ingested into the Bronze layer.

5. Silver Layer — Data Cleaning and Transformation

The Silver layer improves data quality by standardizing text, handling missing values, validating records, and removing duplicate flight IDs.

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW silver_airline_flights
AS
SELECT
  flight_id,
  TRIM(flight_number) AS flight_number,
  COALESCE(
    NULLIF(INITCAP(LOWER(TRIM(airline))), ''),
    'Unknown'
  ) AS airline,
  aircraft_id,
  TRIM(aircraft_type) AS aircraft_type,
  flight_date,
  day_of_week,
  INITCAP(LOWER(TRIM(origin_city))) AS origin_city,
  INITCAP(LOWER(TRIM(destination_city))) AS destination_city,
  origin_region,
  distance_km,
  scheduled_departure,
  actual_departure,
  delay_minutes,
  COALESCE(NULLIF(TRIM(delay_reason), ''), 'Unknown') AS delay_reason,
  seat_capacity,
  passengers,
  avg_fare,
  COALESCE(ticket_revenue, passengers * avg_fare) AS ticket_revenue,
  fuel_cost,
  source_file,
  loaded_at
FROM (
  SELECT *,
    ROW_NUMBER() OVER (
      PARTITION BY flight_id
```

```
        ORDER BY loaded_at DESC
      ) AS rn
  FROM bronze_airline_flights
  WHERE flight_id IS NOT NULL
    AND flight_date IS NOT NULL
    AND passengers <= seat_capacity
    AND passengers >= 0
    AND avg_fare >= 0
) cleaned
WHERE rn = 1;
```

Verify Silver Records:

```
SELECT COUNT(*) AS silver_count
FROM workspace.flights.silver_airline_flights;
```

Output:

MetricResultSilver records1,321

What this tells us: The Silver layer contains 1,321 cleaned and deduplicated records.

6. Rejected Records

Records that fail the defined validation conditions are captured separately.

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW rejected_airline_flights
AS
SELECT *
FROM bronze_airline_flights
WHERE flight_id IS NULL
  OR flight_date IS NULL
  OR passengers > seat_capacity
  OR passengers < 0
  OR avg_fare < 0;
```

Verify Rejected Records:

```
SELECT COUNT(*) AS rejected_count
FROM workspace.flights.rejected_airline_flights;
```

Output:

MetricResultRejected records19

7. Gold Layer — Business Analytics

I created five Gold tables to support business-level analysis.

7.1 Monthly Flight Summary

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW gold_monthly_flights
AS
SELECT
  DATE_FORMAT(flight_date, 'yyyy-MM') AS flight_month,
  COUNT(*) AS total_flights,
  SUM(passengers) AS total_passengers,
  ROUND(AVG(delay_minutes), 2) AS avg_delay_minutes,
  ROUND(SUM(ticket_revenue), 2) AS total_revenue
FROM silver_airline_flights
GROUP BY DATE_FORMAT(flight_date, 'yyyy-MM');
```

View Results:

```
SELECT *
FROM workspace.flights.gold_monthly_flights
ORDER BY flight_month;
```

Key Results:

- Total monthly groups: 12.
- Total flights: 1,321.
- Total passengers: 181,030.

Business Insight: This table helps analyze monthly flight activity, passenger traffic, average delays, and revenue.

7.2 Airline Performance Summary

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW gold_airline_summary
AS
SELECT
  airline,
  COUNT(*) AS total_flights,
  SUM(passengers) AS total_passengers,
  ROUND(AVG(delay_minutes), 2) AS avg_delay_minutes,
  ROUND(SUM(ticket_revenue), 2) AS total_revenue,
  ROUND(AVG(avg_fare), 2) AS avg_fare
```

```
FROM silver_airline_flights
GROUP BY airline;
```

View Results:

```
SELECT *
FROM workspace.flights.gold_airline_summary
ORDER BY total_flights DESC;
```

Key Results:

AirlineFlightsPassengersSky Jet38550,196Air Bharat32242,396Blue Wing26734,032Fly Easy18624,601Coastal Air15019,294Unknown111,511

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Business Insight: Sky Jet recorded the highest number of flights and passengers among the airlines in the cleaned dataset.

7.3 Route Performance Summary

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW gold_route_summary
AS
SELECT
  origin_city,
  destination_city,
  COUNT(*) AS total_flights,
  SUM(passengers) AS total_passengers,
  ROUND(AVG(delay_minutes), 2) AS avg_delay_minutes,
  ROUND(SUM(ticket_revenue), 2) AS total_revenue
FROM silver_airline_flights
GROUP BY origin_city, destination_city;
```

View Results:

```
SELECT *
FROM workspace.flights.gold_route_summary
ORDER BY total_flights DESC;
```

Business Insight: This table helps identify frequently operated routes and compare passenger traffic, average delays, and revenue across routes.

7.4 Flight Delay Analysis

Flights were categorized into three groups:

- On Time: Delay of 15 minutes or less.
- Minor Delay: Delay greater than 15 and up to 60 minutes.
- Major Delay: Delay greater than 60 minutes.

SQL Query:

```
CREATE OR REFRESH MATERIALIZED VIEW gold_delay_summary
AS
SELECT
  CASE
    WHEN delay_minutes <= 15 THEN 'On Time'
    WHEN delay_minutes <= 60 THEN 'Minor Delay'
    ELSE 'Major Delay'
  END AS delay_category,
  COUNT(*) AS total_flights,
  ROUND(AVG(delay_minutes), 2) AS avg_delay_minutes,
  SUM(passengers) AS total_passengers,
  ROUND(SUM(ticket_revenue), 2) AS total_revenue
FROM silver_airline_flights
GROUP BY
  CASE
    WHEN delay_minutes <= 15 THEN 'On Time'
    WHEN delay_minutes <= 60 THEN 'Minor Delay'
    ELSE 'Major Delay'
  END;
```

View Results:

```
SELECT *
FROM workspace.flights.gold_delay_summary;
```

Actual Results:

Delay CategoryFlightsAverage DelayOn Time8892.56 minMinor Delay28137.47 minMajor Delay151150.49 min

Business Insight: 889 flights were categorized as On Time, while 151 flights experienced Major Delays. Major Delay flights had an average delay of approximately 150.49 minutes.

7.5 Data Quality Summary

SQL Query:

```
CREATE OR REPLACE TABLE workspace.flights.gold_data_quality AS
SELECT
  (SELECT COUNT(*)
   FROM workspace.flights.bronze_airline_flights) AS bronze_count,

  (SELECT COUNT(*)
   FROM workspace.flights.silver_airline_flights) AS silver_count,   (SELECT COUNT(*)
   FROM workspace.flights.rejected_airline_flights) AS rejected_count,   (SELECT COUNT(*)
   FROM workspace.flights.bronze_airline_flights)
  -
  (SELECT COUNT(DISTINCT flight_id)
   FROM workspace.flights.bronze_airline_flights)
  -
  (SELECT COUNT(*)
   FROM workspace.flights.bronze_airline_flights
   WHERE flight_id IS NULL) AS duplicate_count;
```

View Results:

```
SELECT *
FROM workspace.flights.gold_data_quality;
```

Actual Results:

MetricRecordsBronze1,365Silver1,321Rejected19Duplicates25

Business Insight: The data quality table provides a quick overview of ingestion, cleaning, rejected records, and duplicate counts. These are the counts produced by the implemented checks.

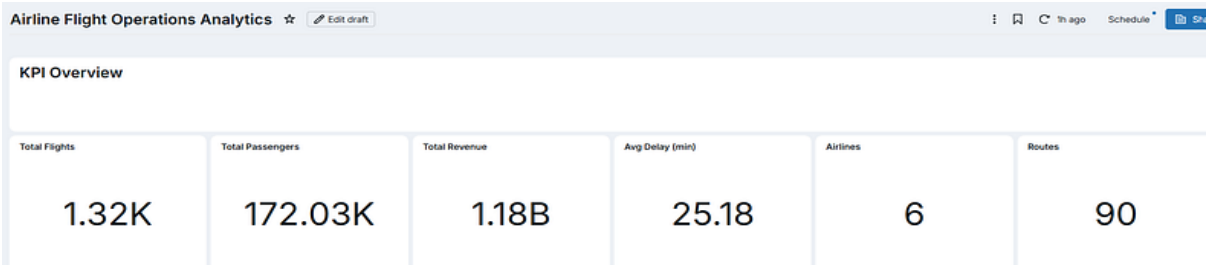
8. Dashboard Development

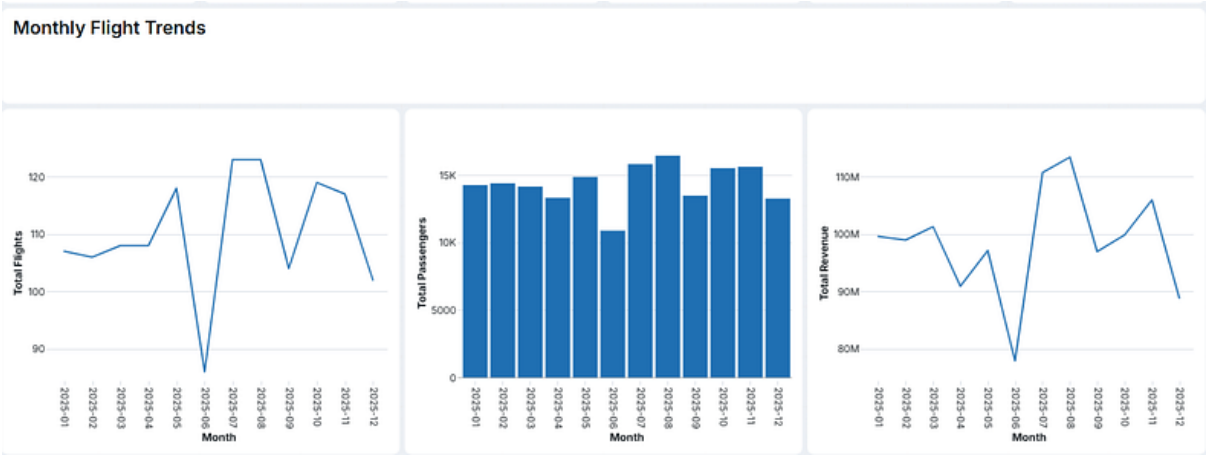
I created an interactive dashboard using Databricks AI/BI Dashboards.

The dashboard includes:

- KPI overview
- Monthly flight trends
- Airline performance
- Route analysis
- Delay analysis
- Data quality overview

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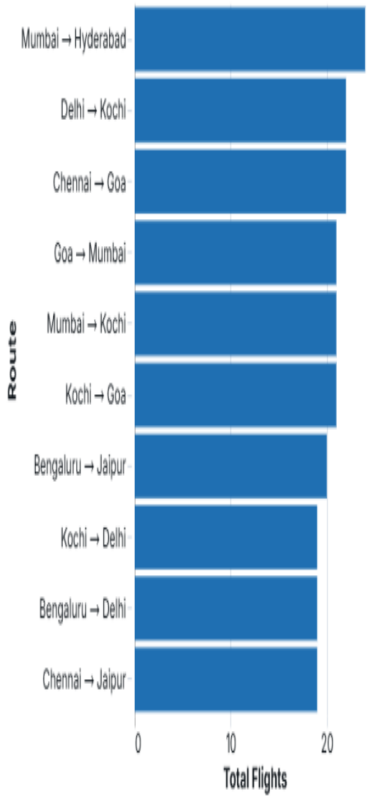


Airline Performance Summary

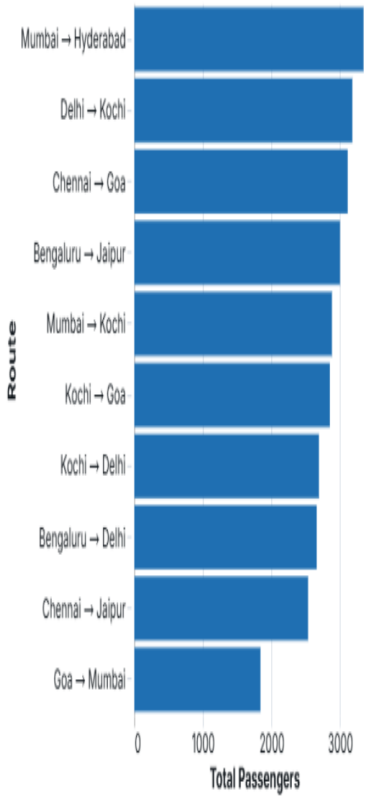
airline	total_flights	total_passengers	avg_delay_minutes	total_revenue	avg_fare
Coastal Air	150	19294	20.21	128818310.00	6336.33
Air Bharat	322	42396	23.08	297326280.00	6727.08
Fly Easy	186	24601	23.34	165914600.00	6546.72
Unknown	11	1511	24.09	10425840.00	6709.09
Sky Jet	385	50196	28.20	345502820.00	6658.62
Blue Wing	267	34032	27.59	233672550.00	6566.70

Route Analysis

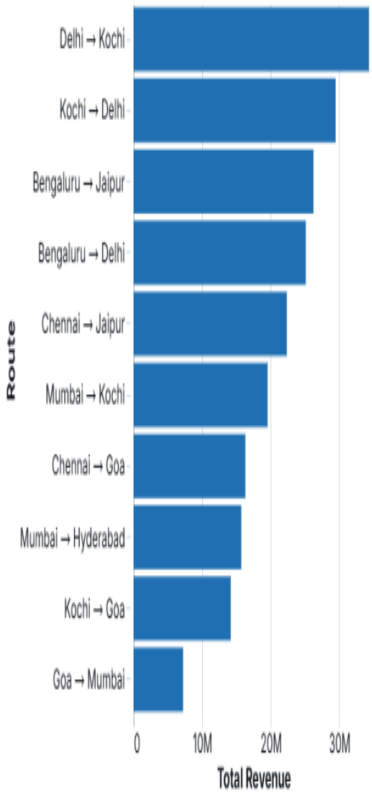
Top 10 Busiest Routes by Flights



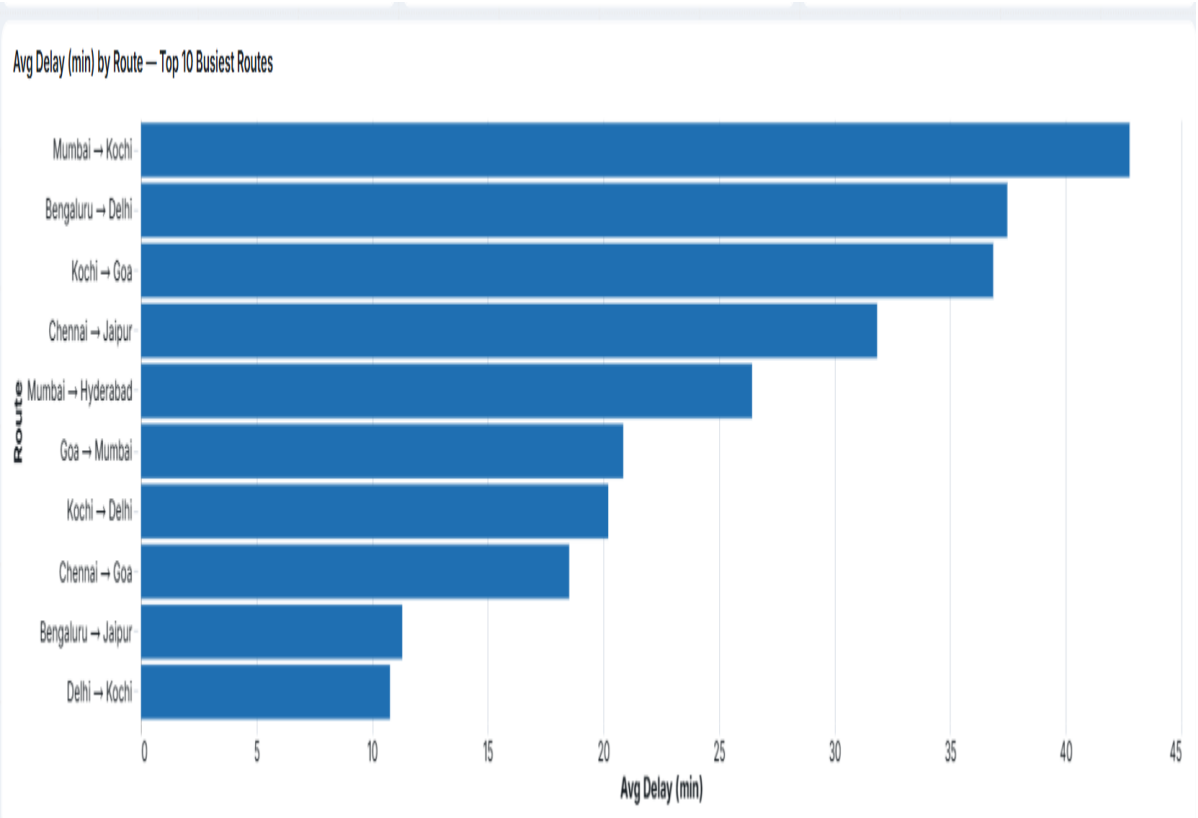
Top 10 Routes by Passengers



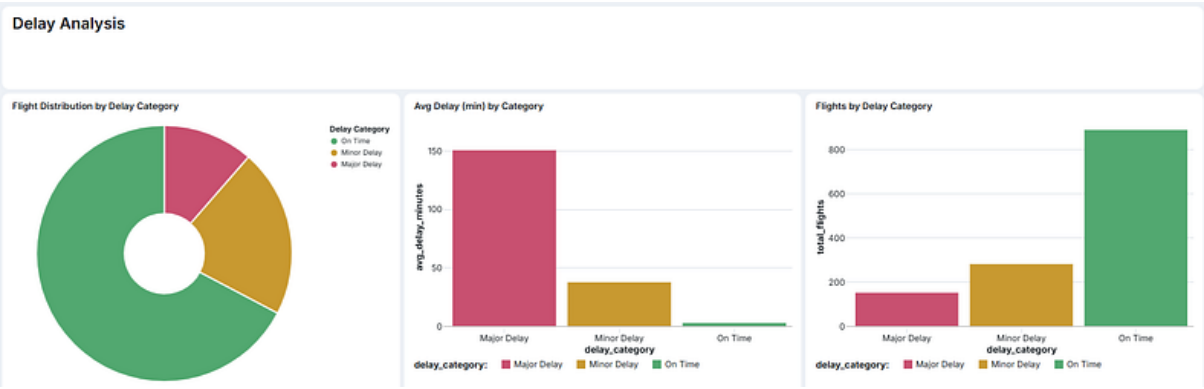
Top 10 Routes by Revenue



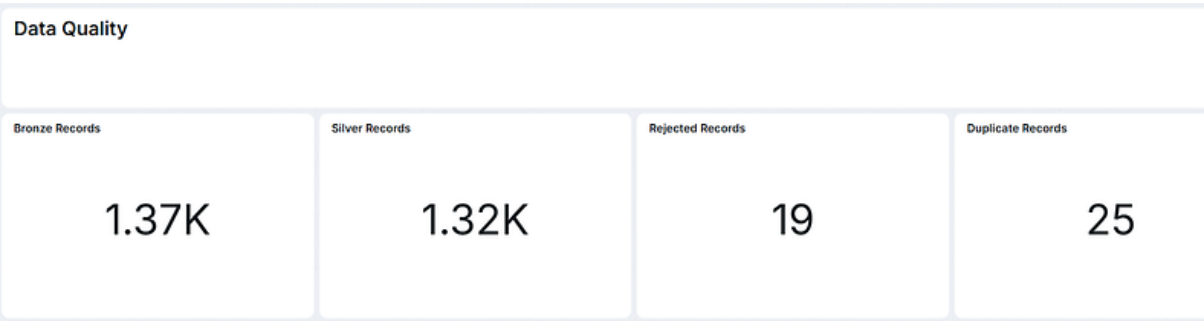
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9. What I Learned

Through this project, I gained practical experience in:

- Data ingestion using Databricks.
- SQL-based data cleaning and transformations.
- Deduplication using ROW_NUMBER().
- Building Bronze, Silver, and Gold layers.
- Creating analytical summary tables.
- Performing data quality checks.
- Developing dashboards for business reporting.

10. Conclusion

This project helped me understand how raw airline operational data can be transformed into useful business insights using Databricks and SQL.

It was a valuable hands-on learning experience in data engineering and analytics.